Key to multiple choice questions:

1.

M^{2+} : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$

M : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8$

Atomic number = 26

Mass number =56

No. of neutrons = 56 -26 = 30

2. $E = h\nu = hc/\lambda$

$$= \frac{6.626\times10^{-34}\ \text{J s}\times3\times10^{8}\ \text{m s}^{-1}}{45\times10^{-9}\ \text{m}}$$

$$= 4.42\times10^{-18}\ \text{J}$$

3.

$$\frac{E1}{E2} = \frac{25\ \text{eV}}{50\ \text{eV}} = \frac{1}{2}$$

$$\frac{hc}{\lambda_1}\times\frac{\lambda_2}{hc} = \frac{1}{2}$$

$$2\lambda_2 = \lambda_1$$

4. Spliting of spectral lines in magnetic field is called Zeeman effect and spliting of spectral lines in electric field, is called Stark effect.

5. **Correct statement:** For n=1, the electron has more negative energy than it does for n=6 which means that the electron is **strongly** bound in the smallest allowed orbit.

6. n = 6 to n=5

$E_6 = -13.6 / 6^2$; $E_5 = -13.6 / 5^2$

$E_6 - E_5 = (-13.6/6^2) - (-13.6/5^2)$

$= 0.166$ eV atom^{-1}

$E_5 - E_4 = (-13.6/5^2) - (-13.6/4^2)$

$= 0.306$ eV atom^{-1}

9. Spin quantum number

For the first electron $m_s = +½$

For the second electron $m_s = -½$

10. Eu : [Xe] $4f^7, 5d^0, 6s^2$

Gd : [Xe] $4f^7, 5d^1, 6s^2$

Tb: [Xe] $4f^9, 5d^0, 6s^2$

11 $2(2l+1) = 4l+2$

12. Orbital angular momentum = $\sqrt{(l(l+1)}\ h/2\pi$ For d orbital = $\sqrt{(2 \times 3)}\ h/2\pi = \sqrt{6}\ h/2\pi$

13 n = 3; l=1; m = -1

either $3p_x$ or $3p_y$

i.e., Maximum two electrons can be accomodated either in $3p_x$ or in $3p_y$

14 No. of radial node = n-l-1

No. of angular node = l

for 3p orbital

No. of angular node = l = 1

No. of radial node = n-l-1 = 3-1-1 = 1

15. n = 3; l = 0; m_l = 0 - one s orbital

n = 3; l = 1; m_l = -1, 0, 1 - three p orbitals

n = 3; l = 2; m_l = -2, -1, 0, 1, 2 - five d Orbitals

Overall nine orbitals are possible.

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